# API Testing

## Introduction

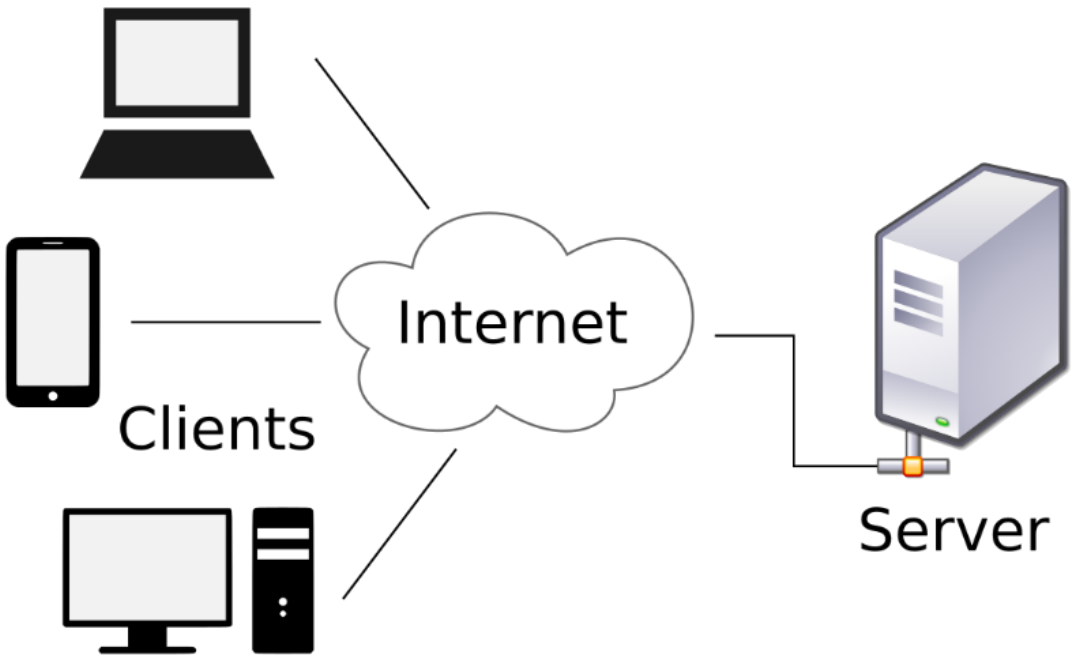
### What is Client & Server?

**Client** is a device or software (like a computer, phone, or browser) that sends requests to get services or data.

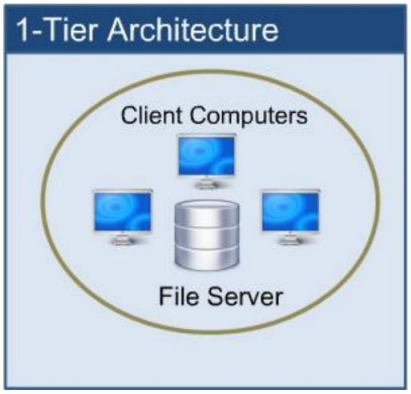
For example, when you use a browser to open a website, your browser acts as the client.

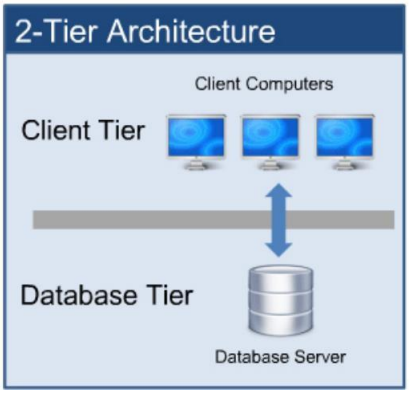
**Server** is a powerful computer or software that receives those requests, processes them, and sends back the requested data or service.

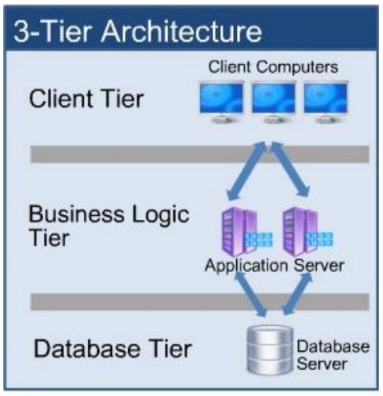
For instance, when you open a website, the server provides the webpage to your browser.

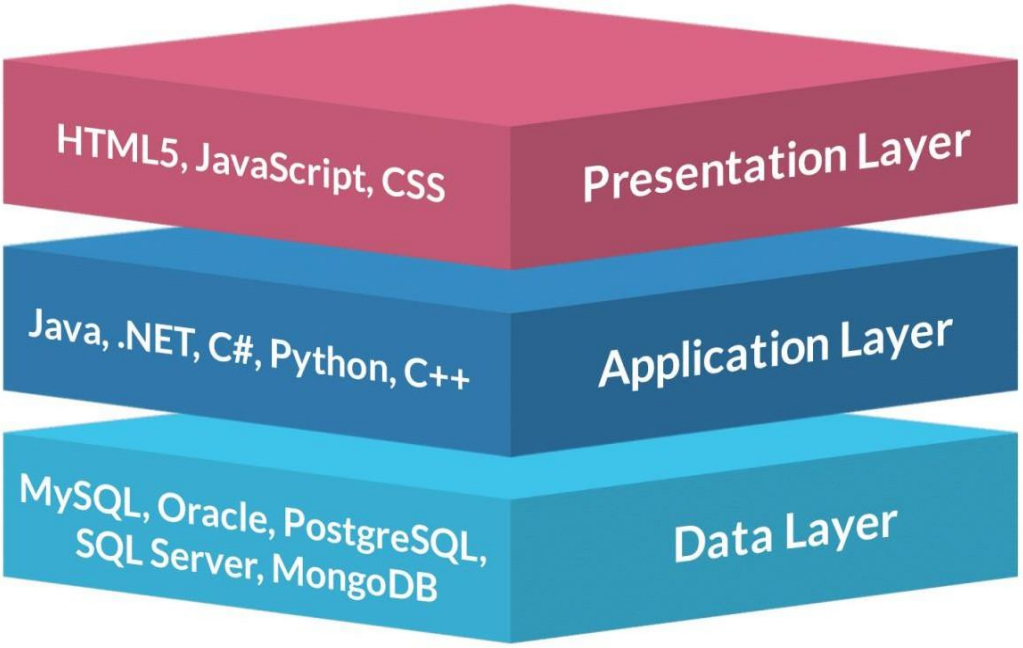


# Client/Server Architecture



# What is an API?

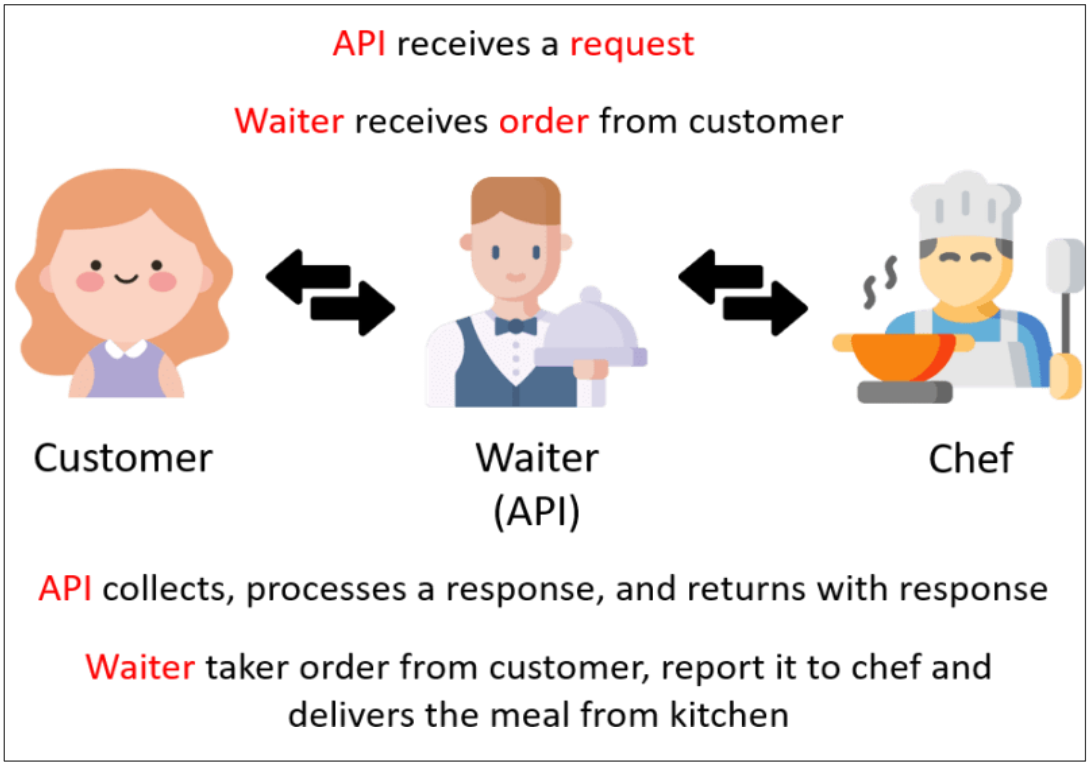
**API (Application Programming Interface)** is like a messenger that helps two programs or systems talk to each other and share information.

**For example:**

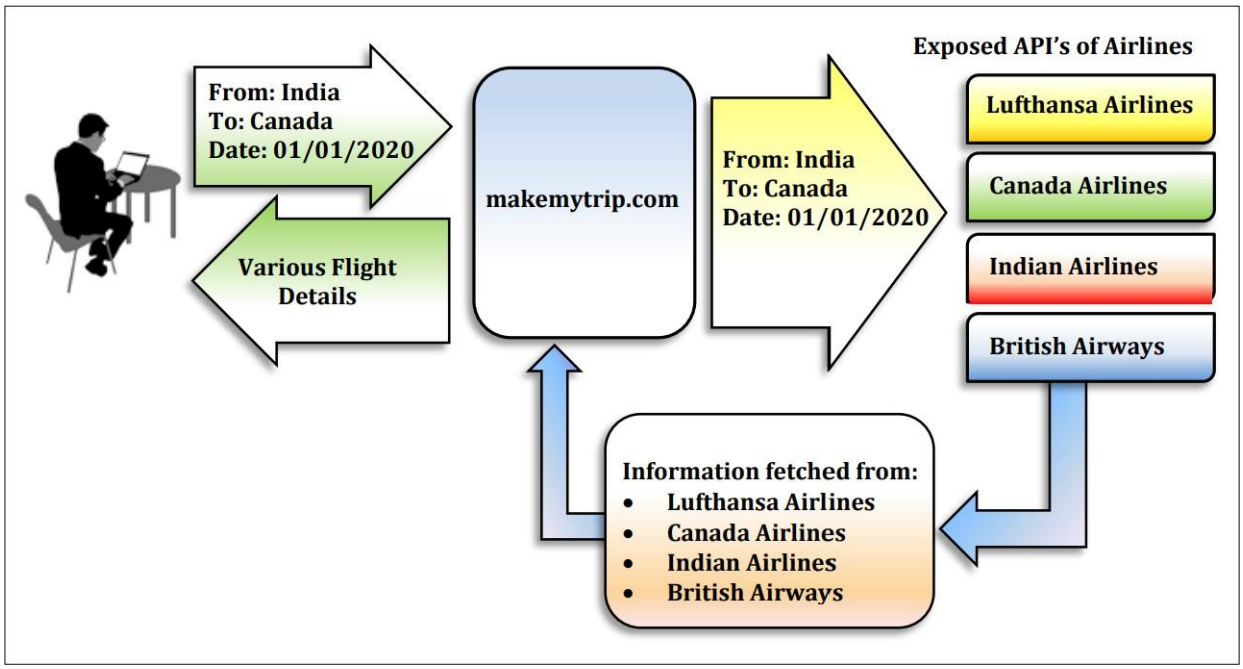
- Imagine you're using a **food delivery app**. When you select a restaurant and place an order, the app uses an API to communicate with the restaurant's system to check the menu, place your order, and confirm it.
- APIs define the rules for how this communication happens, like what data can be sent, how to ask for it, and what the response will look like.

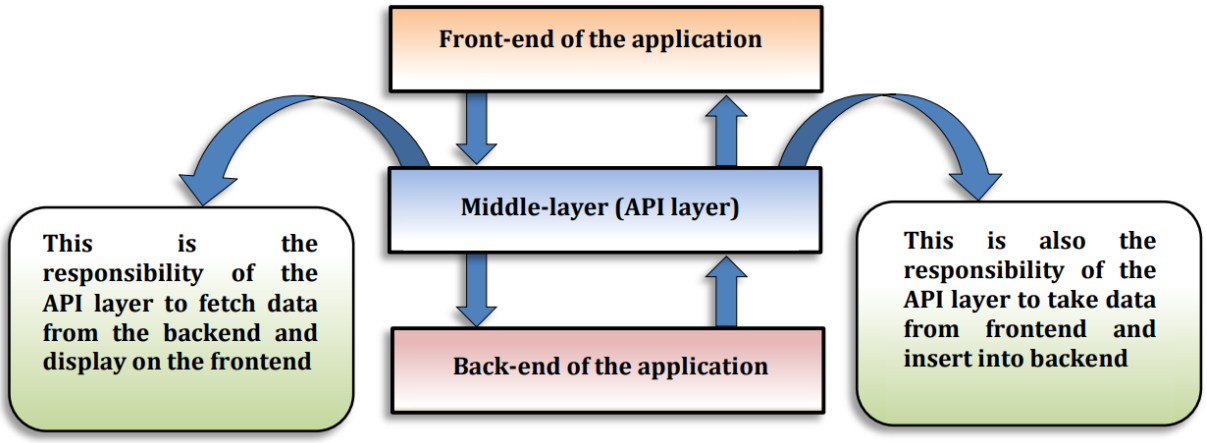
It's **like a waiter in a restaurant**, taking your request to the kitchen and bringing back your food. The API (waiter) ensures both sides (you and the kitchen) understand each other!

# API - Restaurant Analogy



# API - MakeMyTrip Analogy




**** API is a way of communication between two layers.**

# Types Of API

APIs come in different types based on how and where they are used.

### Open APIs (Public APIs):

- These are open for everyone to use. Developers can access them without restrictions (or with minimal requirements).
- Example: Google Maps API lets any app show maps and directions.

### Internal APIs (Private APIs):

- These are used only within a company or organization. They help internal systems or teams communicate securely.
- Example: A company's HR system API to access employee data for payroll processing.

### Partner APIs:

- These are shared with specific partners or businesses. They require special permissions or agreements to use.
- Example: A travel app using an airline's API to show flight details.

### Composite APIs:

- These combine multiple APIs into one call, allowing you to get data from different sources in a single request.
- Example: An e-commerce app using a composite API to retrieve product info, pricing, and customer reviews all at once.

# API Vs Webservice Vs Microservice

Here's a simple comparison of API, Web Service, and Microservice:

### 1. API (Application Programming Interface)

- **What it is:** A way for different software systems or applications to communicate with each other.
- **How it works:** It defines rules for how one app can request data or services from another. APIs don't have to use the internet—they can work locally, too.

- **Example:** Your phone's weather app uses an API to get weather data from a weather server.

## 2. Web Service

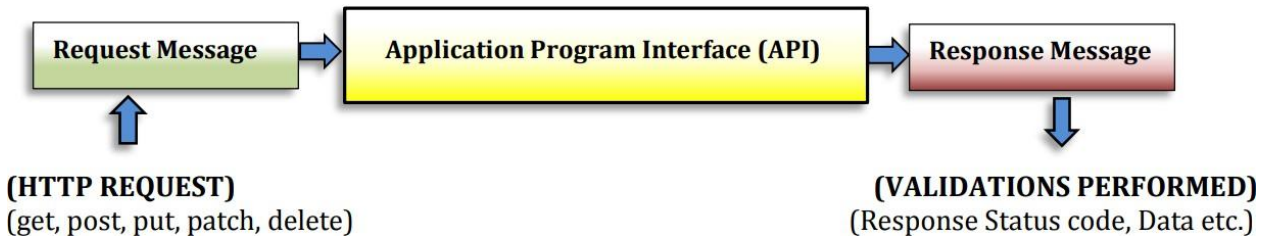
- **What it is:** A type of API that works specifically over the internet (using HTTP).
- **How it works:** It enables two systems to exchange data (often using XML or JSON).
- **Example:** A payment gateway like PayPal's web service allows e-commerce websites to process payments online.

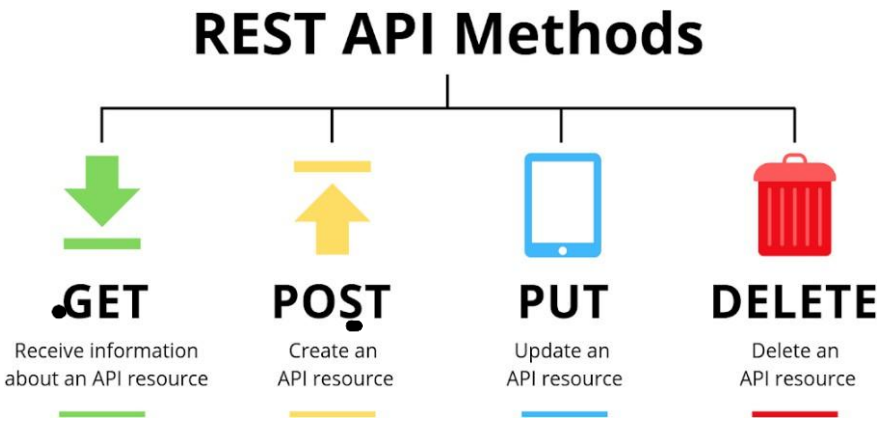
**Key point:** All web services are APIs, but not all APIs are web services.

## 3. Microservice

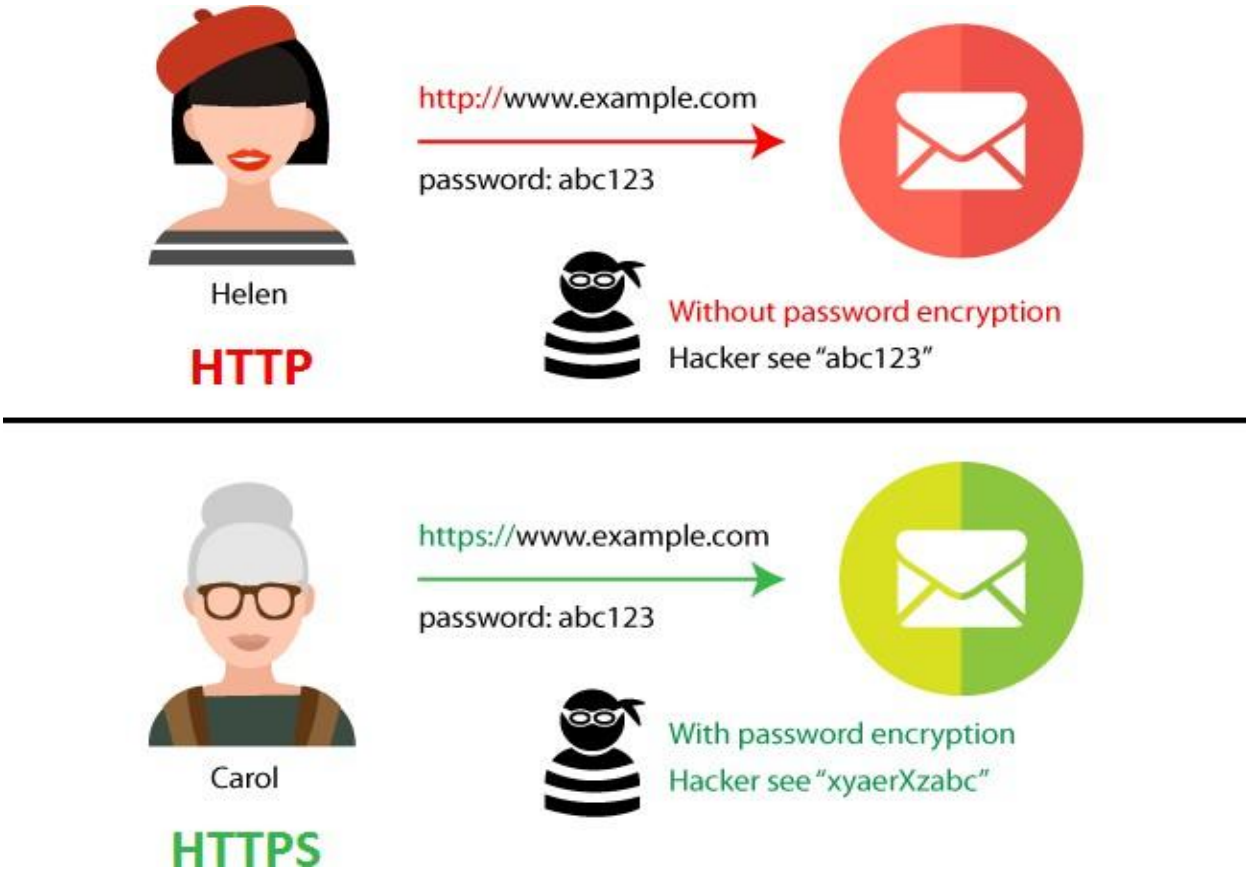
- **What it is:** A small, independent part of an application that does one thing well. A large app is built by combining many microservices.
- **How it works:** Each microservice communicates with other microservices via APIs. They are self-contained and can run independently.
- **Example:** In an online shopping app, separate microservices might handle user login, product catalogue**,** payments, and order tracking.

## Rest API HTTP Methods






## http Vs https

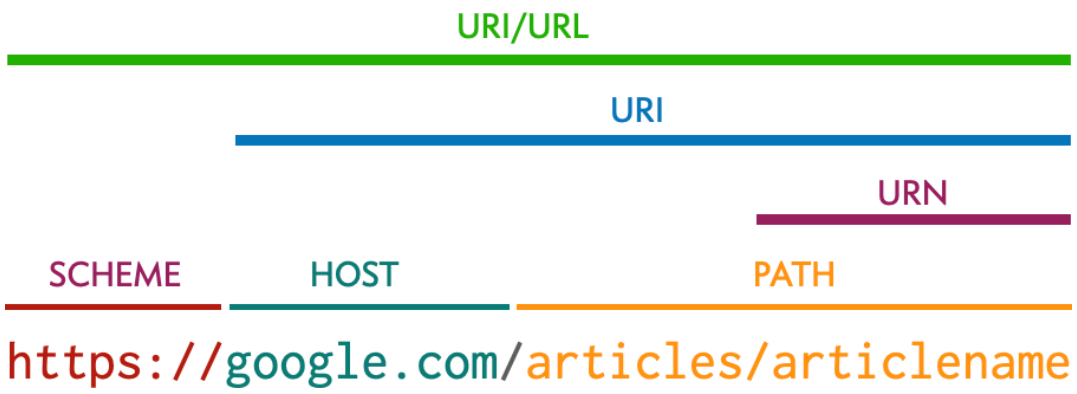


# Terminologies

**URI –** Uniform Resource Identifier

**URL –** Uniform Resource Locator

**URN –** Uniform Resource Name




## Feature & Resource

'**Feature'** is the term used in manual testing to test some functionality and similarly '**Resource'** is the term used in API Automation testing referring some functionality.

## Payload

payload means **body in the HTTP request and response message**.

- Request Payload
- Response Payload

# HTTP Status Codes

| | |
|---|---|
| **1XX** Informational codes | The server has received the request and is currently processing it |
| **2XX** Success codes | The server processed the request after successfully receiving and understanding it |
| **3XX** Redirection codes | The server has received the request, but it is being redirected to another location. In some rare instances, additional actions beyond a redirect may be required |
| **4XX** Client error codes | The server was unable to locate or access the page or website. This issue originates from the site's end |
| **5XX** Server error codes | The client submitted a valid request, but the server was unable to fulfill it |

| | |
|---|---|
| **1XX Information** | |
| 100 | Continue |
| 101 | Switching Protocols |
| 102 | Processing |
| 103 | Early Hints |
| 2XX Success | |
| 200 | OK |
| 201 | Created |
| 202 | Accepted |
| 203 | Non-Authoritative Information |
| 205 | Reset Content |
| 206 | Partial Content |
| 207 | Multi-Status (WebDAV) |
| 208 | Already Reported (WebDAV) |
| 226 | IM Used (HTTP Delta Encoding) |
| **3XX Redirection** | |
| 300 | Multiple Choices |
| 301 | Moved Permanently |
| 302 | Found |
| 303 | See Other |
| 304 | Not Modified |
| 305 | Use Proxy |
| 306 | Unused |
| 307 | Temporary Redirect |
| 308 | Permanent Redirect |
| **4XX Client Error** | |
| 400 | Bad Request |
| 401 | Unauthorized |
| 402 | Payment Required |
| 403 | Forbidden |
| 404 | Not Found |
| 405 | Method Not Allowed |
| 406 | Not Acceptable |

| | |
|---|---|
| **4XX Client (Continue)** | |
| 407 | Proxy Authentication Required |
| 408 | Request Timeout |
| 409 | Conflict |
| 410 | Gone |
| 411 | Length Required |
| 412 | Precondition Failed |
| 413 | Payload Too Large |
| 414 | URI Too Large |
| 415 | Unsupported Media Type |
| 416 | Range Not Satisfiable |
| 417 | Exception Failed |
| 418 | I'm a teapot |
| 421 | Misdirected Request |
| 422 | Unprocessable Entity (WebDAV) |
| 423 | Locked (WebDAV) |
| 424 | Failed Dependency (WebDAV) |
| 425 | Too Early |
| 426 | Upgrade Required |
| 428 | Precondition Required |
| 429 | Too Many Requests |
| 431 | Request Header Fields Too Large |
| 451 | Unavailable for Legal Reasons |
| 499 | Client Closed Request |
| **5XX Server Error Responses** | |
| 500 | Internal Server Error |
| 501 | Not Implemented |
| 502 | Bad Gateway |
| 503 | Service Unavailable |
| 504 | Gateway Timeout |
| 505 | HTTP Version Not Supported |
| 507 | Insufficient Storage (WebDAV) |
| 508 | Loop Detected (WebDAV) |
| 510 | Not Extended |
| 511 | Network Authentication Required |
| 599 | Network Connect Timeout Error |